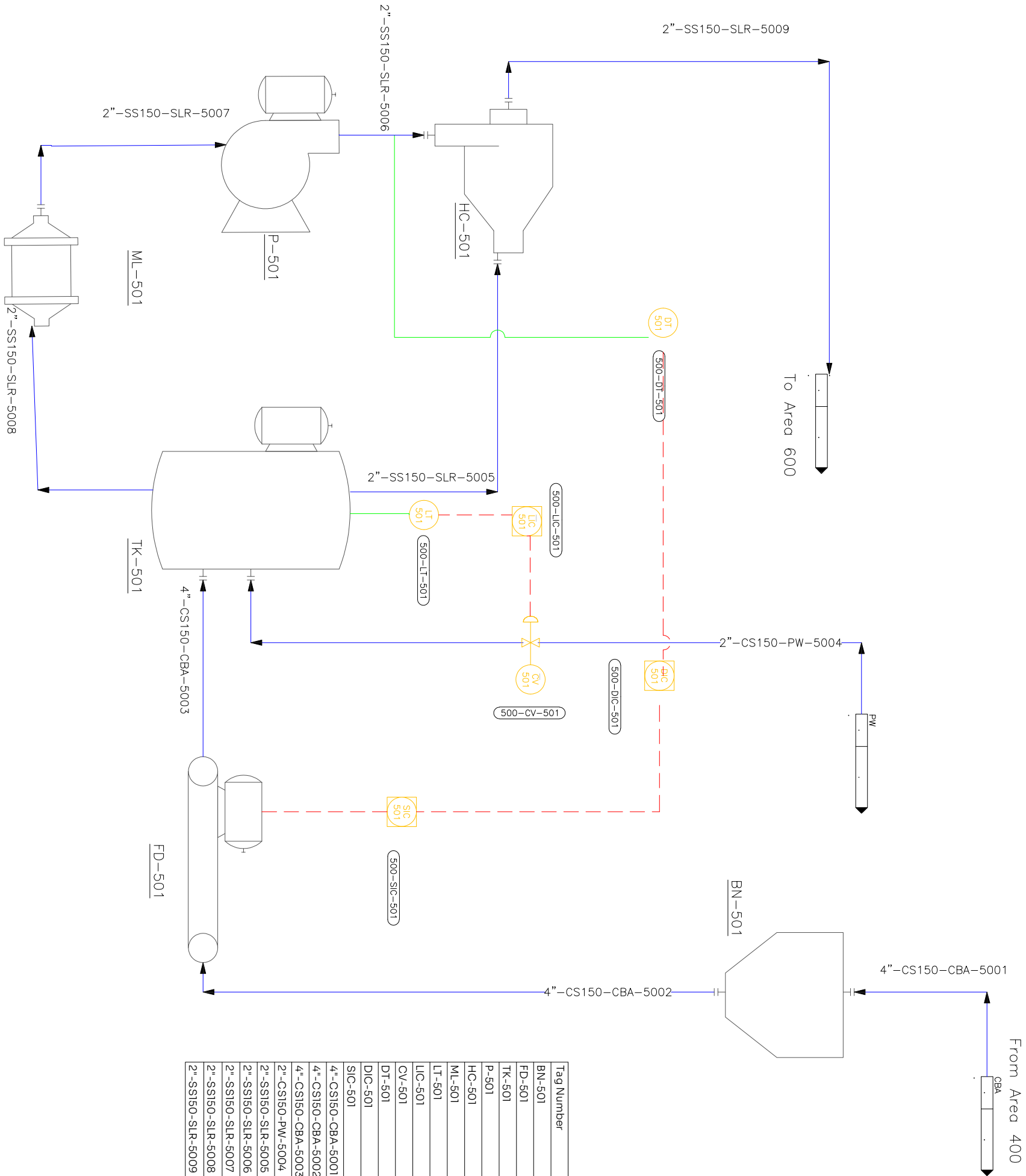




Tag Number	Category	Description
BN-501	Mechanical Equipment	Feed Bin
FD-501	Mechanical Equipment	Weigh Feeder
TK-501	Mechanical Equipment	Slurry Mixing Tank
P-501	Mechanical Equipment	Centrifugal Slurry Pump
HC-501	Mechanical Equipment	Hydrocyclone
ML-501	Mechanical Equipment	Wet Grinding Mill
LT-501	Instrumentation	Level Transmitter
LIC-501	Instrumentation	Level Indicating Controller
DT-501	Instrumentation	Control Valve
DIC-501	Instrumentation	Density Transmitter
SIC-501	Instrumentation	Density Indicating Controller
4"-CS150-CBA-5001	Piping	Speed Indicating Controller
4"-CS150-CBA-5002	Piping	Calclined Bauxite Inlet Line
4"-CS150-CBA-5003	Piping	Calclined Bauxite Feed Line
2"-CS150-PW-5004	Piping	Process Water Line
2"-SS150-SLR-5005	Piping	Slurry Pump Suction Line
2"-SS150-SLR-5006	Piping	Slurry Pump Discharge Line
2"-SS150-SLR-5007	Piping	Hydrocyclone Underflow Line
2"-SS150-SLR-5008	Piping	Mill Discharge Line
2"-SS150-SLR-5009	Piping	Hydrocyclone Overflow Line

General Notes

- All pipes, fittings, and valves shall be designed in accordance with ASME B31.3 process piping standards.
- All instrumentation symbols and nomenclature conform to ISA 5.1 standards.
- Fail-safe actions for all pneumatic control valves
- (fail-open/fail-closed) shall be determined during the HAZOP review phase.
- All slurry piping (SLR) to be designed to maintain a minimum suspension velocity of 1.5 m/s to prevent solid settling.
- Long radius sweeps (5D) shall be used in place of standard 90-degree elbows on all slurry lines to minimize erosive wear.
- Centrifugal slurry pumps (P-501, P-502) require tungsten carbide mechanical seals and flush water tie-ins.

No.

Revision/Issue

Date

Drawing Name

Area500
Wet Grinding and
Classification

Project Name and Address

PAC_P&ID
Final Year Project
PMB, UPO, KNUST, Ghana
Kumasi Ashanti Region

Area

500

Slump

Drawing Number

5

Author

GROUP 7B